

## SMITHERS, BC, Canada

WMO: 719500

Lat: 54.8242N

Lon: 127.1894W

Elev: 1716

StdP: 13.81

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |       |       |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD<br>to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|-------|-------|-----|------|-----------------------|------|------|------|--------------------------|------|-------|
|               |            |      | 99.6%                         |     |       | 99%   |     |      | 0.4%                  |      | 1%   |      |                          |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB  | DP    | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                     | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)   | (g)   | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                      | (o)  | (p)   |
| 1             | -13.3      | -6.7 | -20.9                         | 1.9 | -11.9 | -14.1 | 2.7 | -5.1 | 18.3                  | 28.8 | 16.4 | 26.7 | 2.4                      | 150  | 0.621 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 21.5                   | 82.0            | 61.5 | 77.6 | 59.8 | 73.6 | 57.9 | 63.2                | 77.1 | 61.2 | 74.3 | 59.5 | 70.8 | 5.4                  | 330  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 58.1                            | 76.9 | 66.4 | 56.3 | 71.8 | 64.4 | 54.7 | 67.7 | 63.0 | 29.4          | 77.2 | 28.0 | 74.5 | 26.8 | 71.2 | 75.6              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |     |                    |      | n-Year Return Period Values of Extreme Temperature |     |            |      |            |      |            |      |       |      |
|-------------------|------|------|----------------------------|-----|--------------------|------|----------------------------------------------------|-----|------------|------|------------|------|------------|------|-------|------|
|                   |      |      | Mean                       |     | Standard Deviation |      | n=5 years                                          |     | n=10 years |      | n=20 years |      | n=50 years |      |       |      |
| 1%                | 2.5% | 5%   | Min                        | Max | Min                | Max  | Min                                                | Max | Min        | Max  | Min        | Max  | Min        | Max  |       |      |
| (a)               | (b)  | (c)  | (e)                        | (f) | (g)                | (h)  | (i)                                                | (j) | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |       |      |
| (4)               | 15.2 | 12.6 | 11.4                       | DB  | -19.5              | 88.1 | 7.3                                                | 4.2 | -24.7      | 91.1 | -29.0      | 93.6 | -33.1      | 95.9 | -38.5 | 99.0 |
| (5)               |      |      |                            | WB  | -19.8              | 67.1 | 7.2                                                | 2.8 | -25.0      | 69.2 | -29.3      | 70.8 | -33.3      | 72.4 | -38.6 | 74.4 |

## Monthly Climatic Design Conditions

|                                                                               |                                                                               | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |
|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                               |                                                                               | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                           | DBAvg                                                                         | 39.8   | 20.0  | 23.3  | 31.4  | 40.6  | 49.4  | 55.5  | 59.5  | 58.5  | 50.4  | 40.0  | 28.6  | 19.8  |      |
|                                                                               | DBStd                                                                         | 16.40  | 13.02 | 10.61 | 8.80  | 5.37  | 6.16  | 5.58  | 5.28  | 5.33  | 5.67  | 5.59  | 8.51  | 11.36 |      |
|                                                                               | HDD50                                                                         | 4592   | 929   | 747   | 578   | 286   | 87    | 10    | 1     | 2     | 61    | 314   | 641   | 936   |      |
|                                                                               | HDD65                                                                         | 9215   | 1394  | 1167  | 1043  | 733   | 483   | 290   | 186   | 214   | 437   | 776   | 1091  | 1401  |      |
|                                                                               | CDD50                                                                         | 885    | 0     | 0     | 0     | 3     | 70    | 175   | 296   | 264   | 74    | 3     | 0     | 0     |      |
|                                                                               | CDD65                                                                         | 31     | 0     | 0     | 0     | 0     | 0     | 5     | 15    | 11    | 0     | 0     | 0     | 0     |      |
|                                                                               | CDH74                                                                         | 816    | 0     | 0     | 0     | 1     | 27    | 149   | 333   | 286   | 20    | 0     | 0     | 0     |      |
|                                                                               | CDH80                                                                         | 202    | 0     | 0     | 0     | 0     | 2     | 33    | 97    | 68    | 2     | 0     | 0     | 0     |      |
| Wind                                                                          | WSAvg                                                                         | 3.7    | 3.9   | 3.4   | 4.3   | 4.4   | 4.0   | 3.7   | 3.3   | 3.0   | 3.0   | 3.5   | 3.7   | 3.7   |      |
| Precipitation                                                                 | PrecAvg                                                                       | 21.2   | 2.4   | 1.3   | 1.0   | 0.8   | 1.4   | 1.7   | 1.9   | 1.8   | 2.2   | 2.6   | 2.2   | 2.0   |      |
|                                                                               | PrecMax                                                                       | 29.2   | 4.9   | 2.8   | 2.7   | 1.7   | 3.0   | 3.5   | 4.7   | 3.8   | 3.4   | 4.7   | 4.7   | 4.3   |      |
|                                                                               | PrecMin                                                                       | 14.6   | 0.6   | 0.2   | 0.3   | 0.2   | 0.3   | 0.4   | 0.4   | 0.2   | 0.3   | 1.1   | 0.3   | 0.3   |      |
|                                                                               | PrecStd                                                                       | 2.8    | 1.3   | 0.6   | 0.6   | 0.4   | 0.7   | 0.9   | 1.0   | 0.9   | 0.8   | 0.9   | 1.0   | 0.9   |      |
| Monthly Design<br>Dry Bulb and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%                                                                          | DB     | 43.2  | 43.6  | 54.5  | 68.7  | 77.5  | 84.5  | 88.8  | 86.1  | 76.8  | 60.1  | 47.8  | 42.3  |      |
|                                                                               |                                                                               | MCWB   | 38.4  | 37.7  | 43.1  | 50.6  | 55.7  | 61.5  | 63.0  | 62.4  | 59.4  | 51.8  | 42.7  | 37.6  |      |
|                                                                               | 2%                                                                            | DB     | 39.7  | 40.2  | 48.6  | 59.8  | 72.5  | 78.3  | 82.1  | 81.5  | 70.1  | 55.0  | 43.3  | 38.2  |      |
|                                                                               |                                                                               | MCWB   | 36.4  | 35.5  | 39.8  | 45.3  | 53.8  | 59.4  | 61.5  | 61.9  | 56.8  | 48.2  | 40.0  | 35.2  |      |
|                                                                               | 5%                                                                            | DB     | 36.8  | 37.9  | 45.4  | 55.4  | 68.2  | 73.4  | 77.2  | 76.7  | 65.2  | 51.8  | 40.6  | 35.3  |      |
|                                                                               |                                                                               | MCWB   | 34.8  | 34.2  | 37.9  | 43.3  | 52.0  | 57.2  | 60.1  | 60.1  | 54.8  | 46.2  | 38.1  | 33.4  |      |
|                                                                               | 10%                                                                           | DB     | 34.2  | 35.6  | 42.5  | 51.9  | 63.6  | 68.8  | 72.5  | 71.8  | 61.0  | 48.7  | 38.0  | 32.9  |      |
|                                                                               |                                                                               | MCWB   | 32.9  | 32.9  | 36.1  | 41.4  | 50.0  | 55.4  | 58.8  | 58.2  | 52.7  | 44.0  | 36.0  | 31.7  |      |
|                                                                               | Monthly Design<br>Wet Bulb and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%   | WB    | 39.7  | 38.7  | 44.1  | 51.6  | 57.9  | 63.3  | 66.4  | 65.9  | 61.0  | 53.9  | 43.4  | 38.4 |
|                                                                               |                                                                               |        | MCDB  | 42.5  | 42.3  | 53.0  | 66.1  | 73.0  | 79.3  | 82.4  | 80.3  | 73.0  | 58.3  | 46.3  | 41.7 |
| 2%                                                                            |                                                                               | WB     | 36.9  | 36.3  | 40.6  | 47.2  | 54.9  | 60.7  | 63.6  | 62.9  | 58.4  | 49.5  | 40.5  | 35.6  |      |
|                                                                               |                                                                               | MCDB   | 39.0  | 39.0  | 47.0  | 57.0  | 68.2  | 75.9  | 77.0  | 77.1  | 67.5  | 53.7  | 42.6  | 37.6  |      |
| 5%                                                                            |                                                                               | WB     | 34.9  | 34.8  | 38.6  | 44.4  | 53.0  | 58.3  | 61.5  | 60.9  | 56.0  | 46.7  | 38.5  | 33.6  |      |
|                                                                               |                                                                               | MCDB   | 36.6  | 37.2  | 44.4  | 52.9  | 64.9  | 70.4  | 73.8  | 74.3  | 63.2  | 50.6  | 40.4  | 35.1  |      |
| 10%                                                                           |                                                                               | WB     | 32.9  | 33.1  | 36.7  | 42.4  | 51.1  | 56.3  | 59.7  | 59.0  | 53.8  | 44.5  | 36.3  | 31.8  |      |
|                                                                               |                                                                               | MCDB   | 34.2  | 35.3  | 41.5  | 50.0  | 61.7  | 66.8  | 70.5  | 69.6  | 59.5  | 48.0  | 37.8  | 32.9  |      |
| Mean Daily<br>Temperature<br>Range                                            | 5% DB                                                                         | MDBR   | 11.1  | 14.8  | 16.4  | 19.9  | 22.2  | 21.2  | 21.5  | 22.5  | 19.7  | 14.3  | 9.4   | 10.4  |      |
|                                                                               |                                                                               | MCDBR  | 11.6  | 14.1  | 19.6  | 27.3  | 31.6  | 31.2  | 31.9  | 32.6  | 27.9  | 19.3  | 12.0  | 11.4  |      |
|                                                                               |                                                                               | MCWBR  | 9.0   | 10.3  | 12.2  | 15.3  | 15.0  | 14.2  | 13.7  | 14.5  | 15.3  | 12.7  | 9.2   | 9.5   |      |
|                                                                               | 5% WB                                                                         | MCDBR  | 11.1  | 12.7  | 18.7  | 23.4  | 26.8  | 27.5  | 27.3  | 28.9  | 24.3  | 17.2  | 11.3  | 11.0  |      |
|                                                                               |                                                                               | MCWBR  | 8.8   | 9.5   | 12.1  | 14.0  | 13.3  | 13.0  | 12.8  | 13.7  | 13.9  | 11.9  | 8.9   | 9.3   |      |
| Clear-Sky<br>Solar<br>Irradiance                                              | taub                                                                          | 0.268  | 0.277 | 0.304 | 0.356 | 0.374 | 0.378 | 0.363 | 0.359 | 0.324 | 0.311 | 0.277 | 0.247 |       |      |
|                                                                               | taud                                                                          | 2.199  | 2.273 | 2.217 | 2.133 | 2.258 | 2.350 | 2.445 | 2.463 | 2.553 | 2.507 | 2.272 | 2.201 |       |      |
|                                                                               | Ebn at Noon                                                                   | 212    | 258   | 275   | 273   | 275   | 275   | 278   | 272   | 270   | 246   | 207   | 194   |       |      |
|                                                                               | Edh at Noon                                                                   | 19     | 25    | 33    | 42    | 40    | 37    | 33    | 30    | 24    | 20    | 17    | 15    |       |      |
| All-Sky Solar<br>Radiation                                                    | RadAvg                                                                        | 188    | 458   | 828   | 1335  | 1597  | 1750  | 1687  | 1435  | 923   | 484   | 215   | 144   |       |      |
|                                                                               | RadStd                                                                        | 23     | 62    | 81    | 76    | 138   | 167   | 170   | 74    | 94    | 63    | 32    | 19    |       |      |

## Historical Trends

|      |                        | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------|------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|      |                        |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|      |                        | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (46) | Station Only           | N/A   | N/A     | N/A    | +1.80   | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) | Regional (6 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | +106  | N/A   |

Nomenclature: See separate page